Demystifying the population growth model

Mathematical models can seem like magic. Let's see if we cannot gain some intuition for these models by plugging in numbers

The models

Again, we have two equations: The exponential model (left) and the logistic model (right) that describe how the population grows as a function of population size[1], N.

$$\frac{dN}{dt} = rN \qquad\qquad \frac{dN}{dt} = rN\left(1 - \frac{N}{k}\right)$$

We can divide both sides by N to get the *per capita* growth rate[2] in each model

$$\frac{\frac{dN}{dt}}{N} = \frac{rN}{N} \qquad\qquad \frac{\frac{dN}{dt}}{N} = \frac{rN\left(1 - \frac{N}{k}\right)}{N}$$
$$= r \qquad\qquad = r - \frac{r}{k}N$$

Let's choose some values for r and k.

$r =$ _____ (Keep r small, like $0 < r \leq 2$)

$k =$ _____ (Make k big, $400 \leq k \leq 1000$)

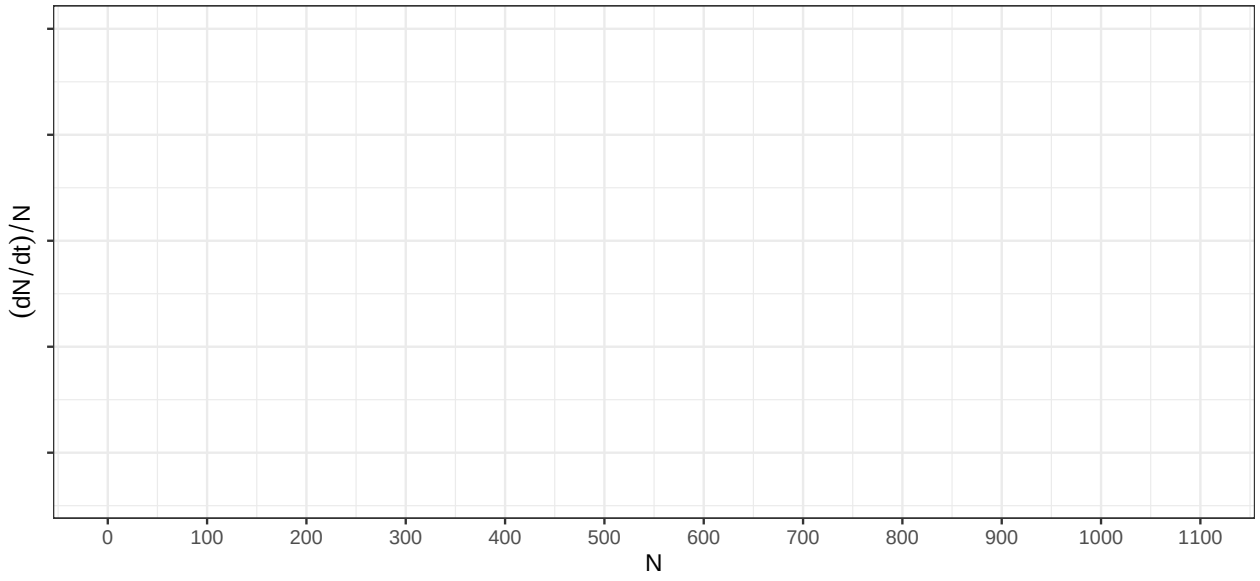
Then fill in the table, below.

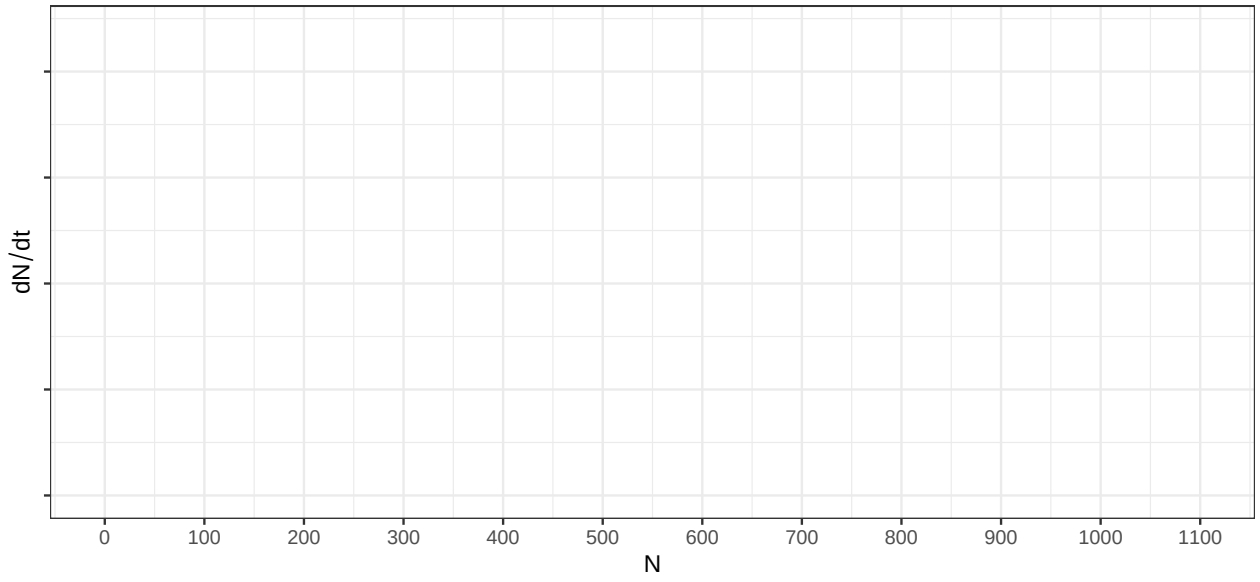
[1] Or population density. It doesn't matter so long as you're consistent

[2] Think of this as the average individual's contribution to population growth rate. If per capita growth rate is zero, each individual just replaces itself.

N	exponential (dN/dt)/N	exponential dN/dt	logistic (dN/dt)/N	logistic dN/dt
0				
100				
200				
300				
400				
500				
600				
700				
800				
900				
1000				
1100				

Then we will graph these numbers…





Where does per capita growth rate equal zero? Where does population growth rate equal zero? Do you understand why?

Next, let's track how the size[3] of a population, $N(t)$, changes over time. Let's fill out the table.

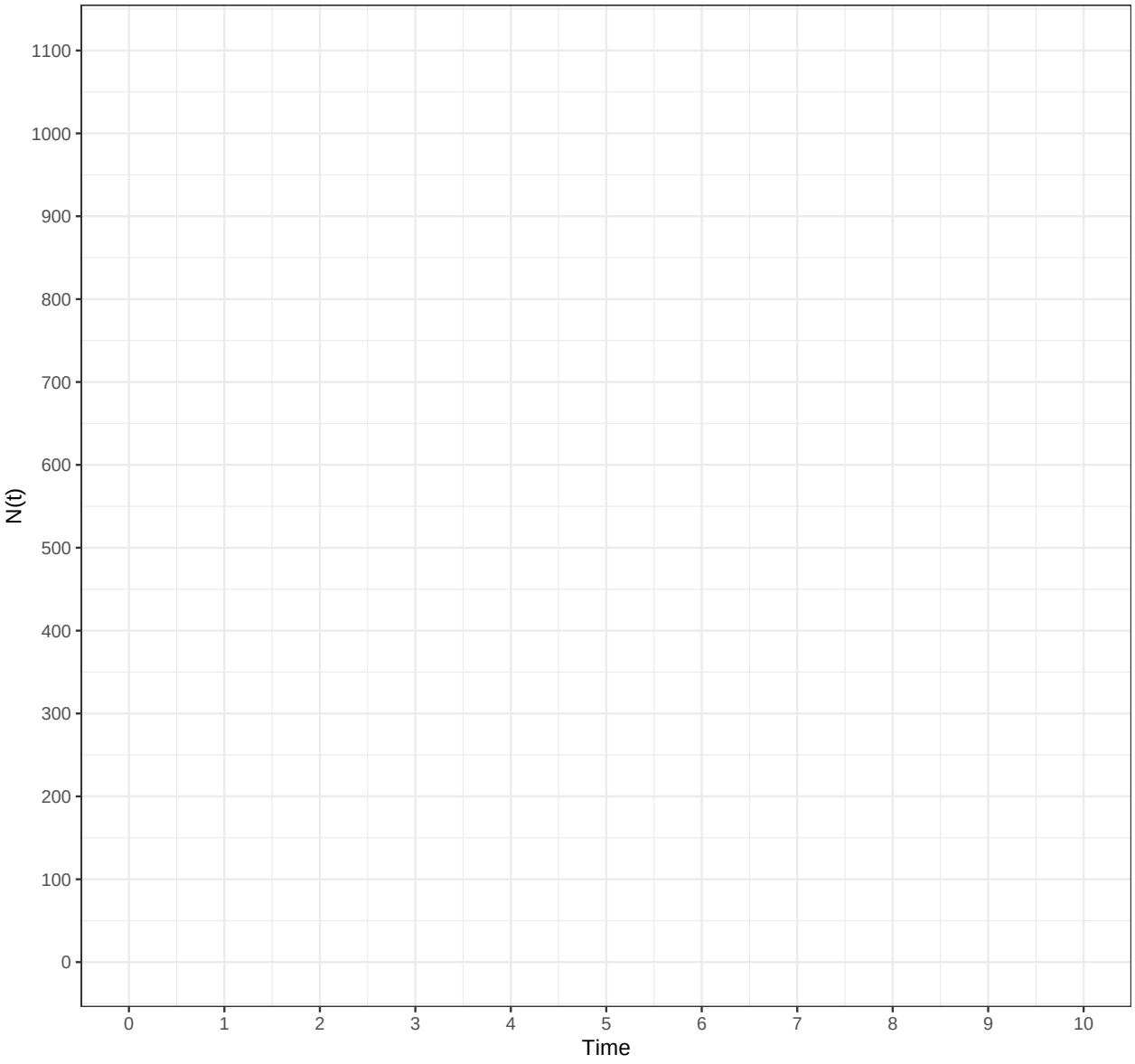
Remember, what we have in the next time step is what we had in *this* time step plus however much the population grows[4].

t	N(t)	dN/dt=rN	N(t)	dN/dt=rN(1-N/K)
0	100		100	
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

And then plot the results

[3] Or density

[4] This is approximately true when the time steps are very small. We are using somewhat course time steps, which means that our answers will diverge from what you would get if you plugged in numbers into the *solutions*, such as \$N(t) = N_0 e^{rt}. We are doing this to learn and understand, rather than for precision, so please do it step by step by hand.


1100
1000
900
800
700
600
500
400
300
200
100
0
N(t)
0
1
2
3
4
5
6
7
8
9
10
Time